

1. REPORT METADATA

Report Generated	2026-08-29 06:13:09 UTC
Source File	test_lease.geojson
Analysis Window	2024-01-01 to 2024-04-30
DEM Source	COPERNICUS/DEM/GLO30_2024_1
Classification	Quad-Lock Multi-Sensor Fusion

2. EXECUTIVE SUMMARY

STATUS: ILLEGAL MINING DETECTED

This report presents the findings of an automated geospatial forensic analysis conducted on the mining lease boundary defined in 'test_lease.geojson'. The analysis utilized multi-sensor satellite data fusion including optical imagery (Sentinel-2), radar data (Sentinel-1 SAR), and topographical data (COPERNICUS/DEM/GLO30_2024_1) to identify and quantify mining activity within and outside the authorized lease boundary.

3. DETECTION METRICS

ILLEGAL MINING (Outside Lease Boundary)

Encroachment Area	10,845,211.85 m2
Encroachment Area	1,084.52 hectares
Excavation Volume	58,852,720.87 m3
Average Pit Depth	5.43 m
Estimated Truckloads (15 m3/truck)	3923514 trucks
Threat Severity	HIGH

LEGAL MINING (Inside Lease Boundary)

Authorized Mining Area	5,397,322.93 m2
Authorized Mining Area	539.73 hectares
Total Excavation Volume	82,128,356.14 m3
Reference Surface Elevation	494.14 m

4. VOLUMETRIC SUMMARY

Category	Area (m2)	Volume (m3)	% of Total
Illegal (Encroachment)	10,845,211.8	58,852,720.9	66.8%
Legal (Authorized)	5,397,322.9	23,275,635.3	33.2%
TOTAL	16,242,534.8	82,128,356.1	100%

5. METHODOLOGY

Lock 1 - Optical Signature (Sentinel-2)

Normalized Difference Built-up Index (NDBI) computed from SWIR and NIR bands to detect bare-soil and disturbed terrain signatures.

Lock 2 - Biological Signature (NDVI)

Normalized Difference Vegetation Index used to filter out vegetated areas. Active mining sites show NDVI below the vegetation threshold.

Lock 3 - Topographical Forensics (DEM)

Copernicus GLO30 DEM with focal-mean smoothing (250m radius) used to reconstruct hypothetical pre-mining terrain. Pits exceeding 2.0m depth flagged as excavation sites.

Fusion Strategy

Triple-Lock verification requires ALL three signatures to co-occur at each pixel. This eliminates false positives from natural rocky terrain, fallow agricultural land, and seasonal vegetation changes.

6. DISCLAIMER

This report is generated by an automated satellite-based monitoring system and is intended for preliminary assessment purposes. The accuracy of results depends on satellite data availability, cloud cover conditions, and the resolution of input DEM data. Field verification is recommended before initiating enforcement actions. All timestamps are in UTC. This document should be treated as CONFIDENTIAL and handled in accordance with applicable data protection regulations.